

VantageScore 2.0: A New Version for a New World of Risk

April 2011

OVERVIEW

VantageScore® is a generic credit risk score launched in March 2006 as a joint venture among the three major credit reporting companies (CRCs)—Equifax, Experian and TransUnion. VantageScore marks the first time the three companies joined forces by combining analytic capabilities and innovative data management techniques with an intuitive scoring scale to produce a model where the same algorithm is utilized by all three CRCs. As determined by testing conducted each year since its introduction, VantageScore has outperformed the three proprietary generic credit scores of the three credit reporting companies. The strength of this performance is attributed to four drivers:

1. Utilizing consumer behavior data from a relevant timeframe to capture consumer perspective on financial debt management.
2. Credit characteristic definitions that effectively differentiate the broad spectrum of the consumer behaviors.
3. Consumer credit performance was calibrated using a randomly selected account providing more refined risk assessment among consumers currently delinquent.
4. A robust segmentation design that creates homogeneous consumer populations for calibrating 12 unique scorecards in the VantageScore model. Two scorecards (high and low risk) are dedicated to consumers who have previously filed bankruptcy, two scorecards (high and low risk) are dedicated to consumers with less than three tradelines on their credit file (defined as “thin file” consumers), and eight scorecards are dedicated to consumers with full credit files, (broken into four risk tiers with two scorecards in each of those tiers).

An additional unique and core strength of the model is the ability to deliver a highly consistent score for any given consumer when their credit information is sourced from different CRCs. A consumer’s credit score can vary, sometimes significantly. This result occurs for two primary reasons:

1. At the time that the credit score is requested, the content of the consumer’s credit file can be different at each CRC.

These differences happen because some lenders may not report data on the consumer to all CRCs or may report data to different CRCs at different times in the month.

2. The model that calculates the credit score is different at each CRC.

Each CRC stores consumer credit file data in slightly different ways. Model developers design algorithms that take these data variances into account in order to obtain optimal predictive performance. For this reason, some model developers install a

OVERVIEW (Cont.)

different algorithm at each CRC which can lead to a different credit score for a consumer at each CRC.

VantageScore development efforts standardized the definitions of the credit characteristics VantageScore considers as closely as possible for each CRC's data in order to remove these variances in the result. This process of redefining credit characteristics to be as consistent as possible is known as "characteristic leveling." Having standardized definitions allows the identical algorithm to be installed at each CRC, removing a significant portion of the variability in a consumer's credit score across the CRCs.

VantageScore Solutions LLC holds the license to the VantageScore algorithm and is responsible for the model's performance. Each year the VantageScore model is validated and the performance results are published to the industry. When appropriate, model developers at the company update the algorithm to ensure that superior performance is maintained.

INTRODUCTION OF VANTAGESCORE 2.0

VantageScore 2.0 was launched in October 2010. The impetus for the updated algorithm came from a substantial shift in the credit economy, most notably observed in the real estate industry from 2007 to 2009. Increasing risk levels across consumer credit tiers, combined with a greater diversity of mortgage product types and consumer behavioral responses motivated VantageScore Solutions to evaluate opportunities for increasing the performance of the VantageScore algorithm.

The evaluation process determined that the underlying architecture of the model, i.e. the leveled characteristic design and segmentation structure, had remained robust. However, performance improvement opportunities were identified from enhancing the underlying consumer behavior data set used to configure the algorithm. Consequently, a dataset was designed that reflected consumer behaviors from multiple outcome periods over an extended timeframe, 2006 to 2009, in order to capture both non-recession and recession-period behaviors. Additionally, 45 million credit files were used to create the dataset—15 million from each CRC. This broad and deep dataset provides an expansive sample of diverse consumer behaviors.

SUMMARY OF PERFORMANCE LIFT WITH VANTAGESCORE 2.0

- VantageScore 2.0 delivers predictive performance lift over the CRC scores and VantageScore 1.0 for all primary industries where credit scores are used in both originations and portfolio management.
- VantageScore 2.0 was created using data blended from two different timeframes from the most recent lending environment: 2006-2008 and 2007-2009. Using a development sample from this extended window captures both a broad and recent set of consumer behaviors, including activity prior to the economic crisis. This reduces algorithm sensitivity to highly volatile behavior that may be observed in a single timeframe and thereby extends performance stability.
- VantageScore 2.0 provides lenders with nearly identical risk assessment across all three CRCs.

DEFINING ALGORITHM PERFORMANCE

Performance with the new version of the algorithm was measured along three dimensions:

1. Score Accuracy
2. Score Consistency
3. Score Stability

SCORE ACCURACY

Traditionally, score accuracy has been the lending industry's sole measure of credit score model performance. Statistical tests such as Gini or Kolmogorov-Smirnov (K-S) measure the effectiveness of the model to separate good performance from bad performance, defined as consumers with debts that are 90 or more days past due. These tests assess a model's effectiveness and indicate performance by assigning values between 0 and 100. A value of 100 indicates that the model perfectly identifies every consumer with good performance and every consumer with bad performance. A value of 0 indicates that the model randomly identifies good and bad performance; in other words the model has no more predictive power than a simple guess. Effective credit score models typically have K-S values between 45 and 70 for the U.S. population.

A second measure of accuracy is the ability of the model to rank order consumers in a monotonically increasing fashion. To test for rank ordering performance, a population of consumers is first scored and then sorted from high score to low score. Once sorted, the consumers are divided into deciles (10-percent groups), where decile one contains the top 10 percent of consumers—consumers with the highest scores, and decile 10 contains the bottom 10 percent of consumers—those with the lowest scores. For each decile, the percentage of consumers with bad performance is calculated. If the model effectively rank orders, then the percentage of consumers with bad performance in decile 10 will be greater than those in decile nine, the percentage of consumers with bad performance in decile nine will be greater than those in decile eight, and so on. An example taken from VantageScore 2.0 on a consumer real estate¹ population appears in Figure 1.

FIGURE 1
REAL ESTATE NEW ACCOUNTS

VANTAGESCORE 2.0	90+DPD ² BAD RATES	
	INTERVAL	CUMUL.
1	0.22%	0.22%
2	0.49%	0.36%
3	0.54%	0.42%
4	0.85%	0.53%
5	1.63%	0.75%
6	2.33%	1.01%
7	4.24%	1.47%
8	6.37%	2.09%
9	10.01%	2.97%
10	19.77%	4.65%

¹ All consumer loans secured by real estate.

² DPD=Days Past Due

**DEFINING
ALGORITHM
PERFORMANCE**

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SCORE CONSISTENCY

Score Consistency can also be defined on two dimensions: Consumer Score Consistency and Risk Alignment.

Consumer Score Consistency is the likelihood that a consumer will receive the same score from two or more different CRCs.

Risk Alignment is the degree to which a credit score reflects the same level of risk at different CRCs. In other words, a VantageScore of 700 denotes the same level of risk, e.g. 2.5 percent, on the U.S. population performance charts generated on different CRC databases. Stronger risk alignment between CRCs provides more accurate risk assessment for the lender and additionally reduces the resources required by the lender for strategy management and score maintenance.

SCORE STABILITY

A final dimension of algorithm performance is defined as the model's ability to maintain its performance over time for both accuracy and consistency.

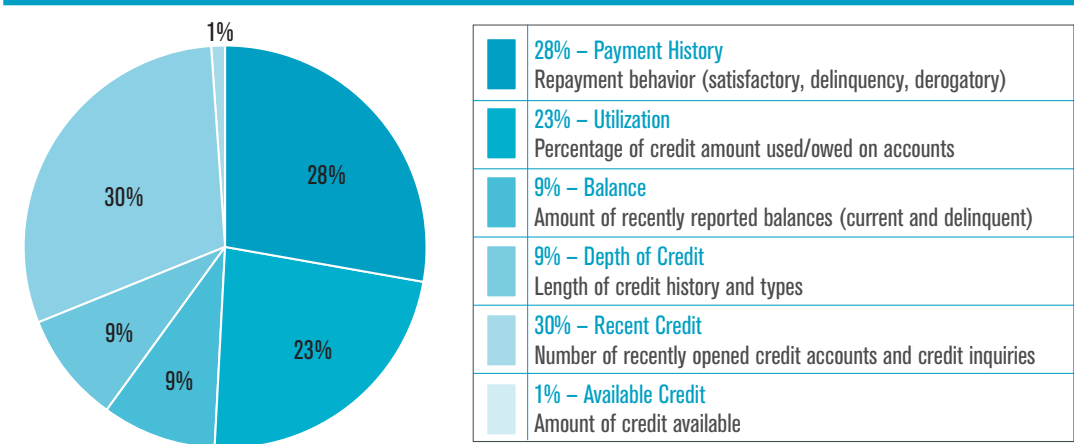
At development, credit score models are optimized on a dataset of specific consumer behavior. As long as consumer behaviors continue to align with those in the dataset used for development, model performance remains at the original level. However as behaviors deviate, as was experienced in the 2007-2009 recession, model performance can deteriorate dramatically. To avoid serious risk exposure, lenders must regularly validate models on their dataset and refine their strategies to compensate for any possible score performance shifts.

If the algorithm performance does not deteriorate or exhibits only minor drops in performance, then lenders utilize significantly fewer resources to maintain their strategies and update internal models. Consequently, scores that remain stable over long periods are highly beneficial to lenders.

THE
VANTAGESCORE 2.0
MODEL

VantageScore 2.0 leverages the core VantageScore platform to deliver improved predictive performance. Several hundred characteristics are created from consumer credit data in order to calculate VantageScore credit scores. These characteristics can be grouped into the six primary categories shown in Figure 2.

FIGURE 2
CONSUMER CHARACTERISTICS CONTRIBUTING
TO A VANTAGESCORE 2.0 CREDIT SCORE



Recent credit behavior, such as inquiries and behavior on newly opened loans, is highly predictive of the consumer’s likelihood to maintain good debt management performance. Payment history and utilization continue to provide significant insight into consumer performance.

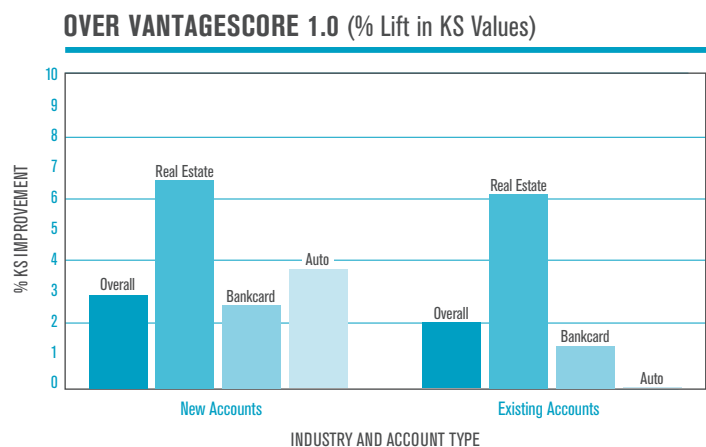
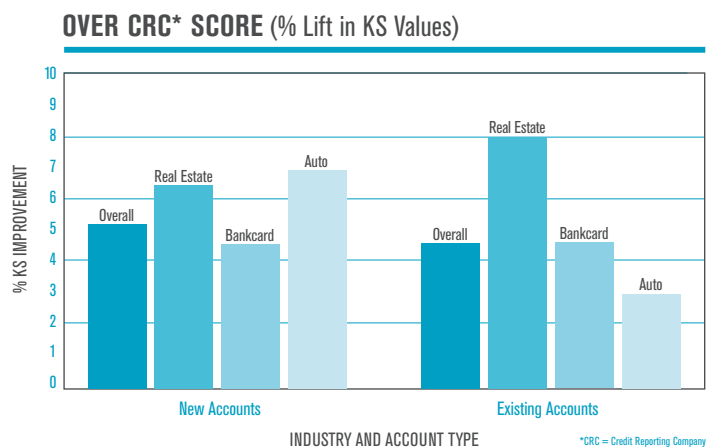
Additional updates contained in the VantageScore 2.0 version include the use of authorized user tradelines in the development dataset and revised adverse action explanations to improve consumer comprehension (adverse action codes remain unchanged).

VANTAGESCORE 2.0 PERFORMANCE: SCORE ACCURACY

ACCURACY HIGHLIGHTS

- VantageScore 2.0 outperforms VantageScore 1.0 and the CRC score in all industries³ and account types.
- VantageScore 2.0 outperforms the benchmark score from each CRC overall by 5 percent for new accounts⁴ and 4.5 percent for existing accounts⁵.
- VantageScore 2.0 outperforms VantageScore 1.0 by 3 percent for new accounts and 2 percent for existing accounts Overall.
- The highest improvement occurs in Real Estate, where VantageScore 2.0 outperforms VantageScore 1.0 by 6.5 percent for new accounts and 6 percent for existing accounts overall.

**FIGURE 3
PERFORMANCE IMPROVEMENT WITH VANTAGESCORE 2.0**



³ "All industries" refers to the main industries where credit scores are used.

⁴ "New accounts" are those accounts opened during the first three months of each performance (study) period.

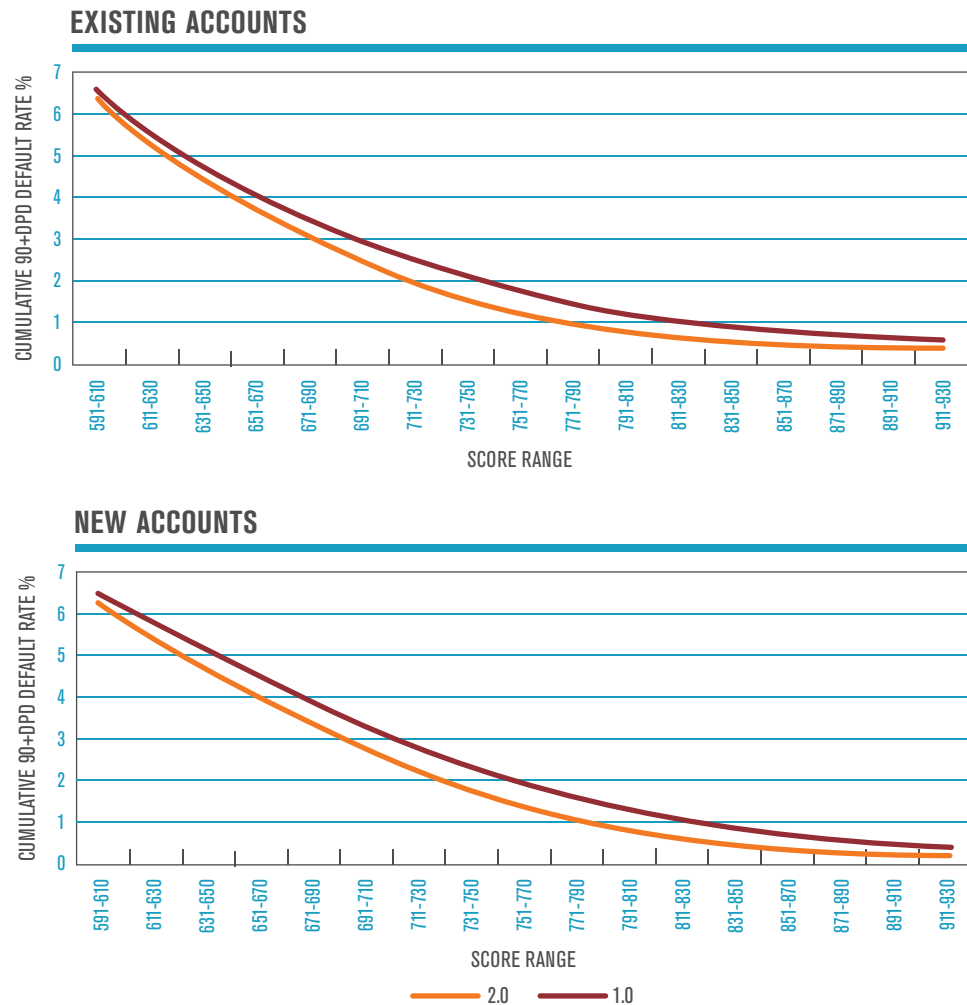
⁵ "Existing accounts" are those opened prior to the performance (study) period.

VANTAGESCORE 2.0 PERFORMANCE: SCORE ACCURACY

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VantageScore 2.0 outperforms both the CRC proprietary credit scores and VantageScore 1.0 across all dimensions—demonstrated above in Figure 3.

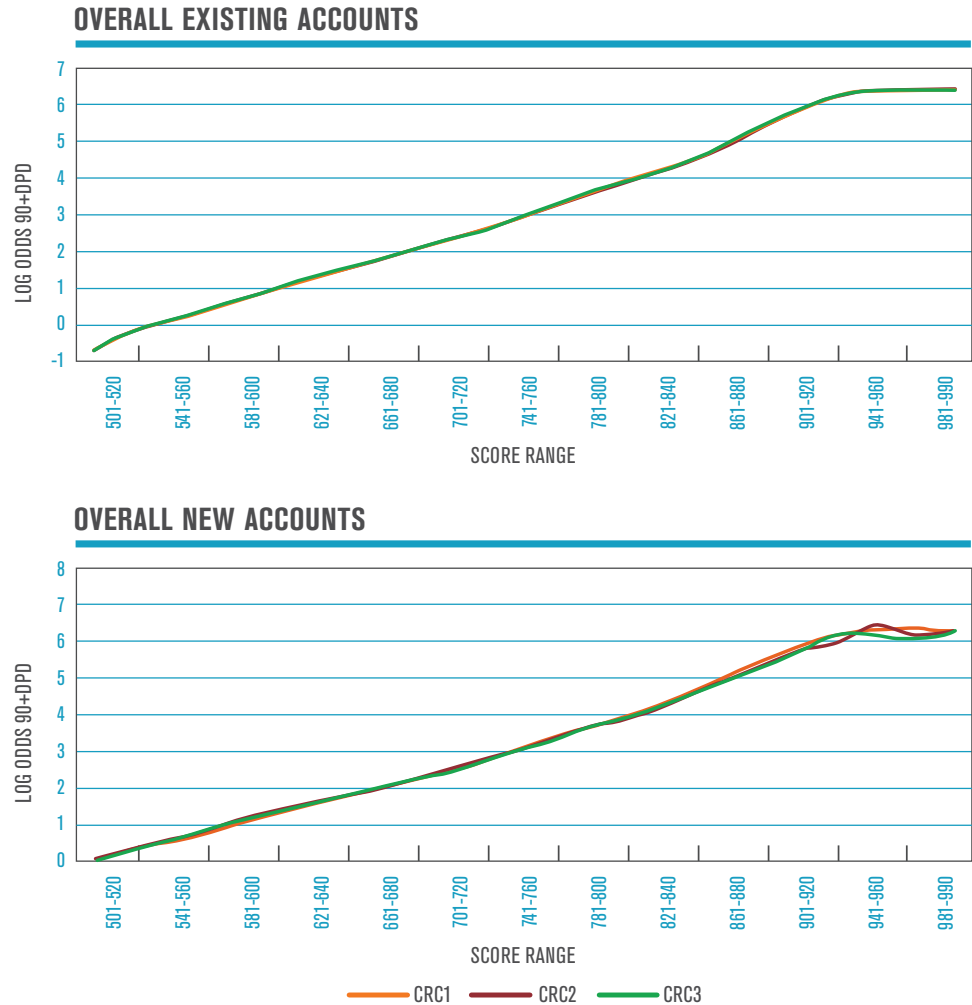
FIGURE 4
90+ DPD RATES: ALL INDUSTRIES



VantageScore 2.0 delivers improved rank ordering capability compared to VantageScore 1.0, as seen in Figure 4.

VANTAGESCORE 2.0 PERFORMANCE: SCORE CONSISTENCY

FIGURE 5
VANTAGESCORE 2.0 ODDS ALIGNMENT



VantageScore 2.0 provides strong risk alignment across the CRCs, with a deviation of less than 1.5 percent difference between risk values at a given score band for existing accounts and 2 percent for new accounts (Figure 5). Similar results are observed for alignment in key industries.

VANTAGESCORE 2.0 PERFORMANCE: SCORE STABILITY

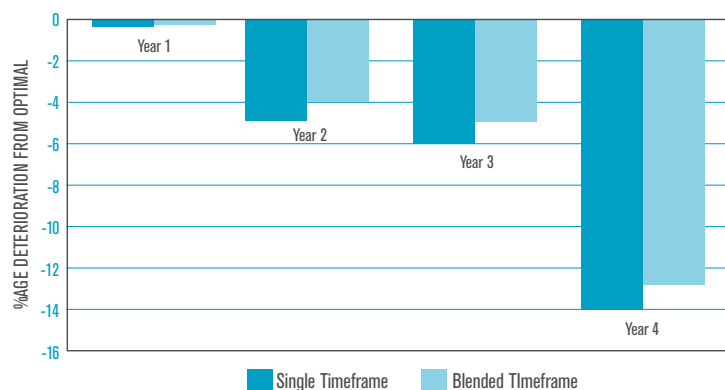
The process for installing a new score within a lender environment is often extensive. The score must first be validated on the lender's portfolio and target universe for effective risk separation. Once authorized, strategies are redesigned and tested then installed within the origination and servicing systems. Lender-developed decision strategies and tools must also be recalibrated to include the new score. Then, the new score performance is typically validated on an annual or biennial basis. Risk, credit, analytic, compliance and technology resources are all required to implement and support these processes. Given this extensive and costly process, lenders are likely to update a score only if the performance has significantly deteriorated.

Score designers can build a degree of performance stability into the algorithm such that performance deteriorates at a lower rate, requiring less score maintenance and update resources for lenders. One approach for creating more stable performance is to develop the algorithm using a dataset obtained from an expanded timeframe, reflecting a greater diversity of financial service products and consumer debt management behaviors.

Typically credit score algorithms are developed on a two-year period of recent history. Consumers in the development dataset are scored at the beginning of the two-year window and their performance is evaluated using the worst status identified throughout the window. VantageScore 2.0 was developed on two consecutive two-year time periods, 2006-2008 and 2007-2009. These multiple timeframes reflect a substantial volume of non-recession and recessionary behaviors. A total of 45 million consumer credit files were used in this process. Consequently, the algorithm effectively interprets consumer risk over a longer continuum of the economic cycle and as a result will deliver stronger predictive performance for a longer period of time than algorithms that are developed on a single timeframe.

A study of predictive performance over time comparing algorithms developed on a single timeframe and those developed on blended timeframes showed that after just one year, the algorithm based on the blended dataset was consistently more predictive in years two through four than the algorithm developed on the single timeframe. Additionally, the performance deterioration using the blended dataset was 16 percent less on average over four years than the single dataset. Figure 6 shows the rate of deterioration for algorithms built on single and blended timeframes over a four-year window.

FIGURE 6
RATE OF DETERIORATION FOR ALGORITHMS BUILT



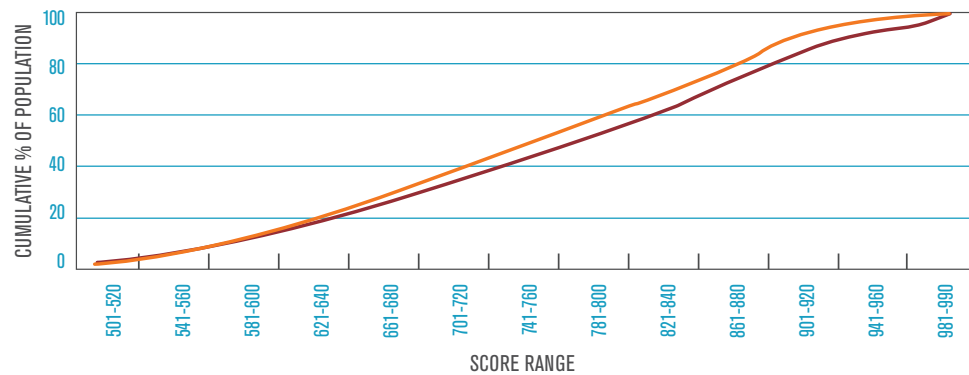
VANTAGESCORE 2.0 PERFORMANCE: SCORE DISTRIBUTION

OVERALL SCORE DISTRIBUTION

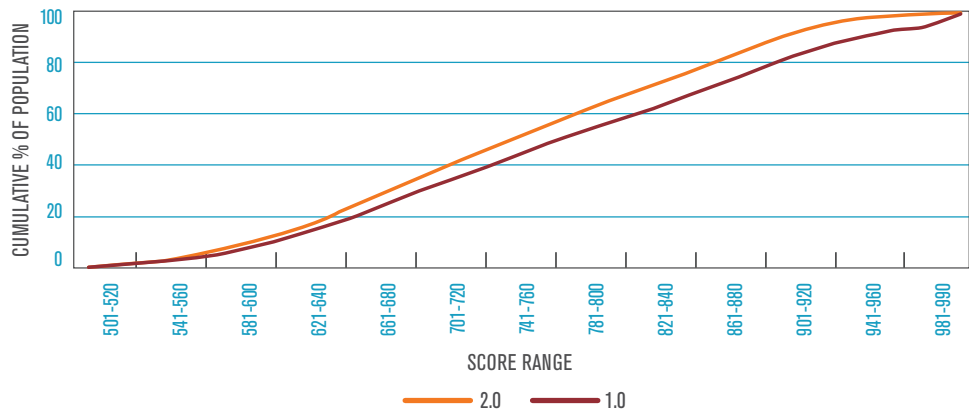
Commensurate with the increased risk in the economy, consumers now score slightly lower reflecting the increased risk, seen below in Figure 7.

FIGURE 7
CUMULATIVE SCORE DISTRIBUTION: VANTAGESCORE 1.0 & VANTAGESCORE 2.0

OVERALL EXISTING ACCOUNTS



OVERALL NEW ACCOUNTS

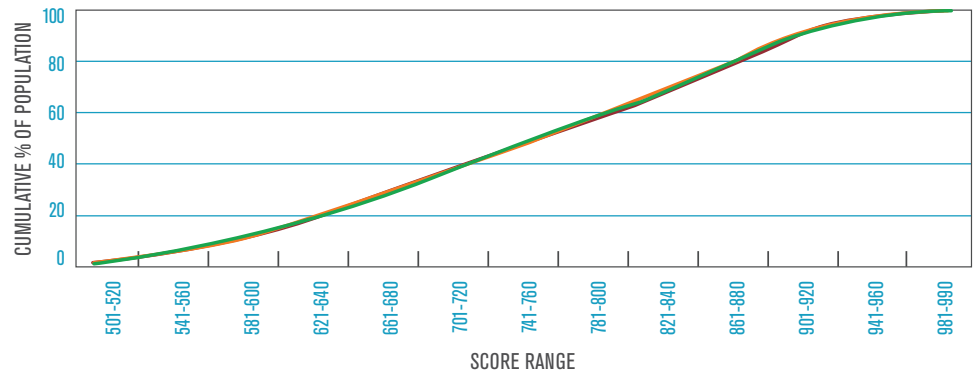


VANTAGESCORE 2.0 PERFORMANCE: SCORE DISTRIBUTION (Cont.)

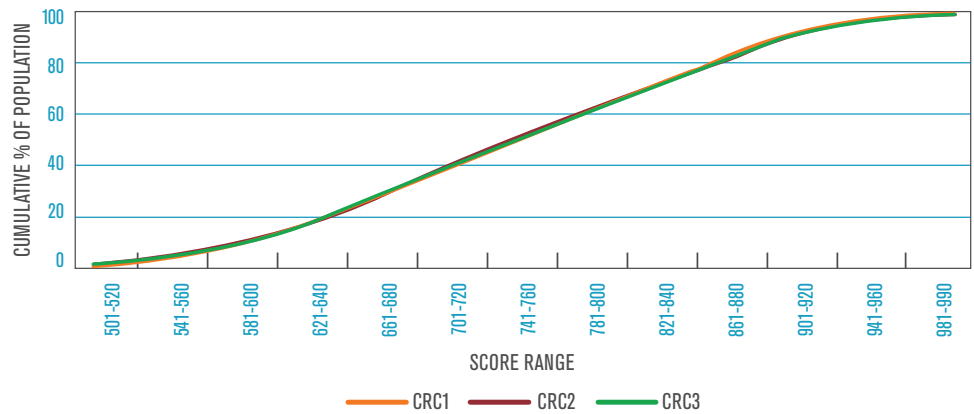
The distribution using VantageScore 2.0 is highly consistent across all three CRCs, as seen in Figure 8 below.

FIGURE 8
VANTAGESCORE 2.0: CONSISTENCY OF CUMULATIVE SCORE DISTRIBUTIONS

ACROSS CRCs FOR OVERALL EXISTING ACCOUNTS



ACROSS CRCs FOR OVERALL NEW ACCOUNTS

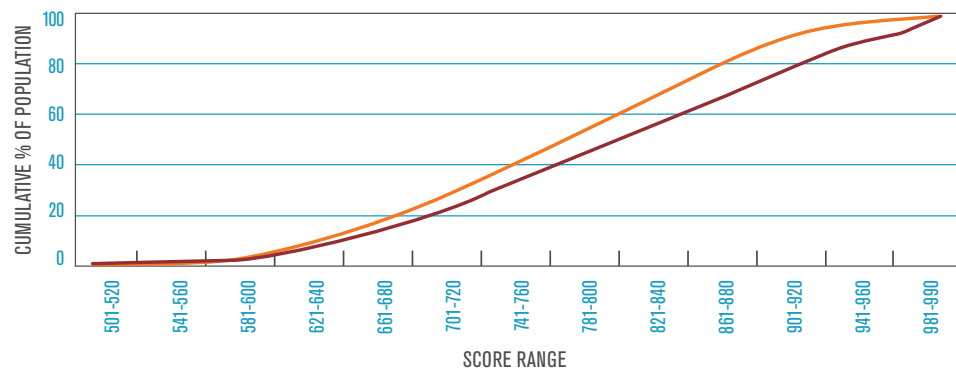


VANTAGESCORE 2.0 PERFORMANCE: SCORE DISTRIBUTION (Cont.)

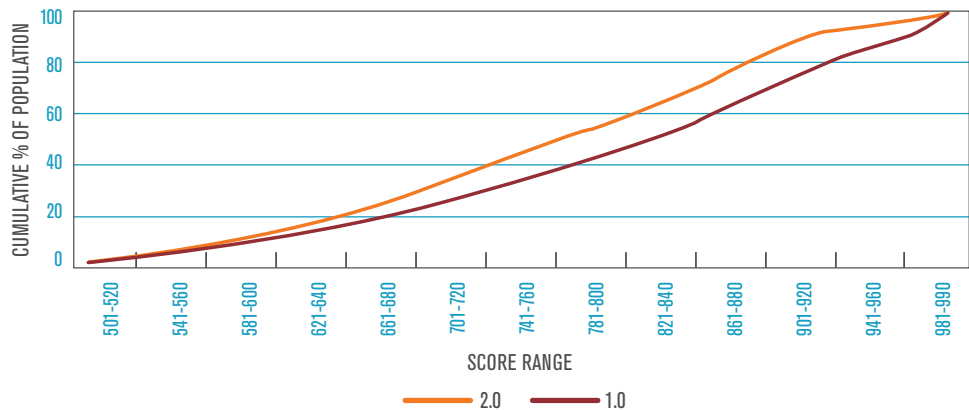
In each industry segment, score distributions concur with the improved performance of VantageScore 2.0, reflecting the increased risk in consumer behaviors captured within the blended development timeframe (Figures 9, 10, 11). Lenders are encouraged to validate VantageScore 2.0 on their own portfolios prior to making policy decisions.

FIGURE 9
CUMULATIVE SCORE DISTRIBUTION: VANTAGESCORE 2.0 & VANTAGESCORE 1.0

REAL ESTATE NEW ACCOUNTS



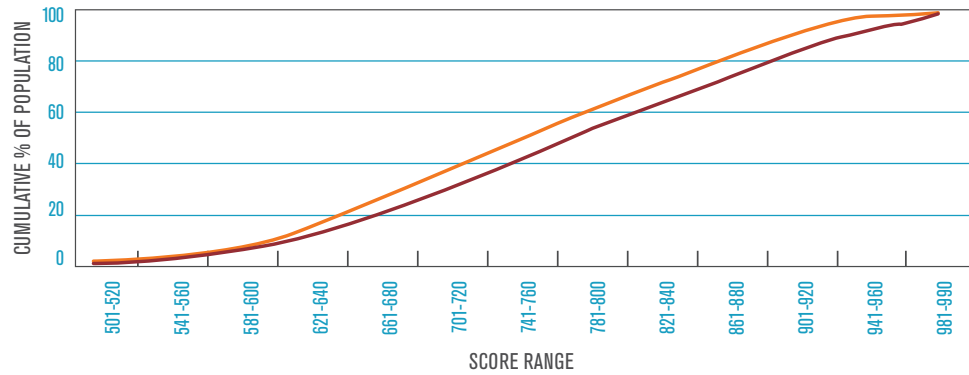
REAL ESTATE EXISTING ACCOUNTS



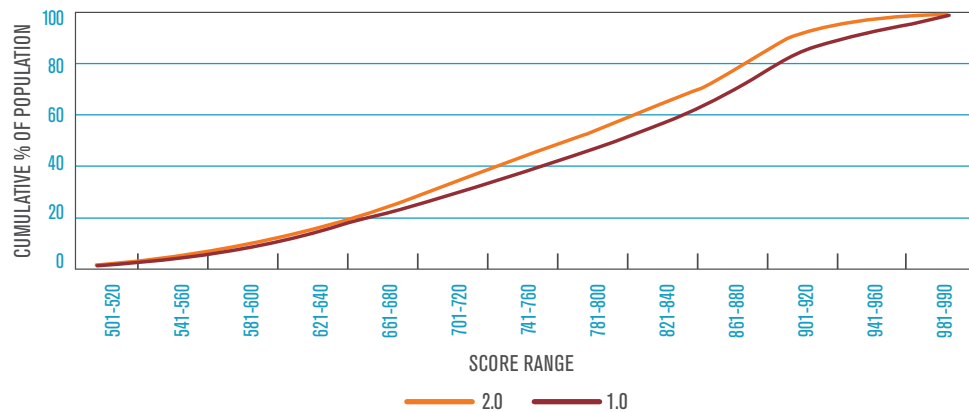
**VANTAGESCORE 2.0
PERFORMANCE:
SCORE
DISTRIBUTION**
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FIGURE 10
CUMULATIVE SCORE DISTRIBUTION: VANTAGESCORE 2.0 & VANTAGESCORE 1.0

BACKCARD NEW ACCOUNTS

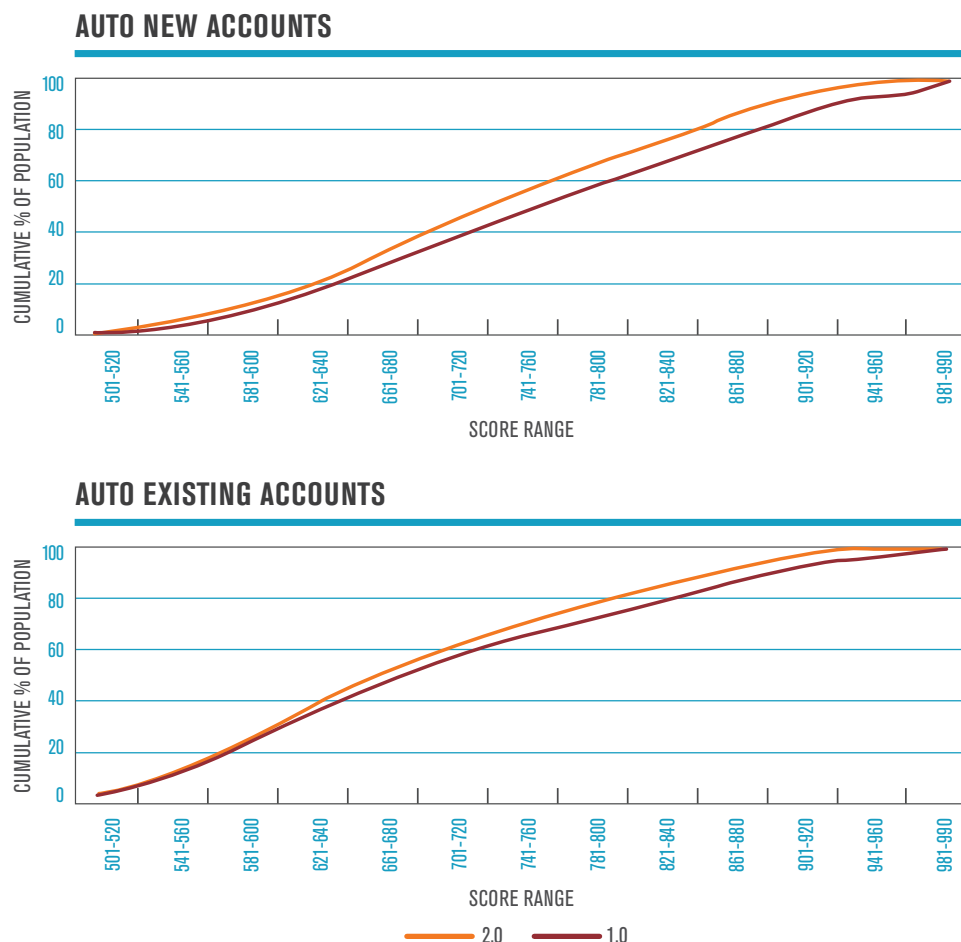


BACKCARD EXISTING ACCOUNTS



VANTAGESCORE 2.0 PERFORMANCE: SCORE DISTRIBUTION (Cont.)

FIGURE 11
CUMULATIVE SCORE DISTRIBUTION: VANTAGESCORE 2.0 & VANTAGESCORE 1.0



SUMMARY

Compelled by changing consumer behavior observed during the volatile economic period from 2007-2009, coupled with new real estate loan products in the years preceding the long recession, VantageScore Solutions introduced a new version of the VantageScore model—VantageScore 2.0. The new version leverages the core VantageScore platform and a development dataset of 45 million consumer credit files over a blended timeframe (2006-2009) to deliver improved predictive performance as well as stability of the model's performance over an extended window.

VantageScore is the only generic scoring model to employ the same characteristics and the same algorithm at each CRC. The result is a more predictive, consistent score for consumers across all three CRCs and more consistent risk assessment for lenders.